SATELLITE INTELLIGENCE PLATFORM

Automated urban damage detection in Ukrainian conflict zones using Sentinel-2 multispectral imagery

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What we built, at a glance.

3

Cities Studied

Mariupol · Kharkiv · Bakhmut

41

Satellite Scenes

Cloud-filtered Sentinel-2 L2A acquisitions spanning 2021 Q1 through 2024 Q2.

0.934

MLP F1 Score

Best model performance across all metrics, with AUC-ROC of 0.980.

// THE PROBLEM

Detect infrastructure damage automatically, without entering the conflict zone.

Timely, scalable assessment across 3 Ukrainian cities, 14 quarterly periods.

WHAT

Classify Sentinel-2 satellite scenes as damaged (Label 1) or undamaged (Label 0) using spectral indices and machine learning.

WHY IT MATTERS

Ground surveys are dangerous in active war zones. Aerial access is restricted. Aid organisations need data now to plan safely.

WHAT MAKES IT HARD

Only 41 scenes total. No pixel-level ground truth. Winter cloud cover. Three cities with distinct destruction patterns and timelines.

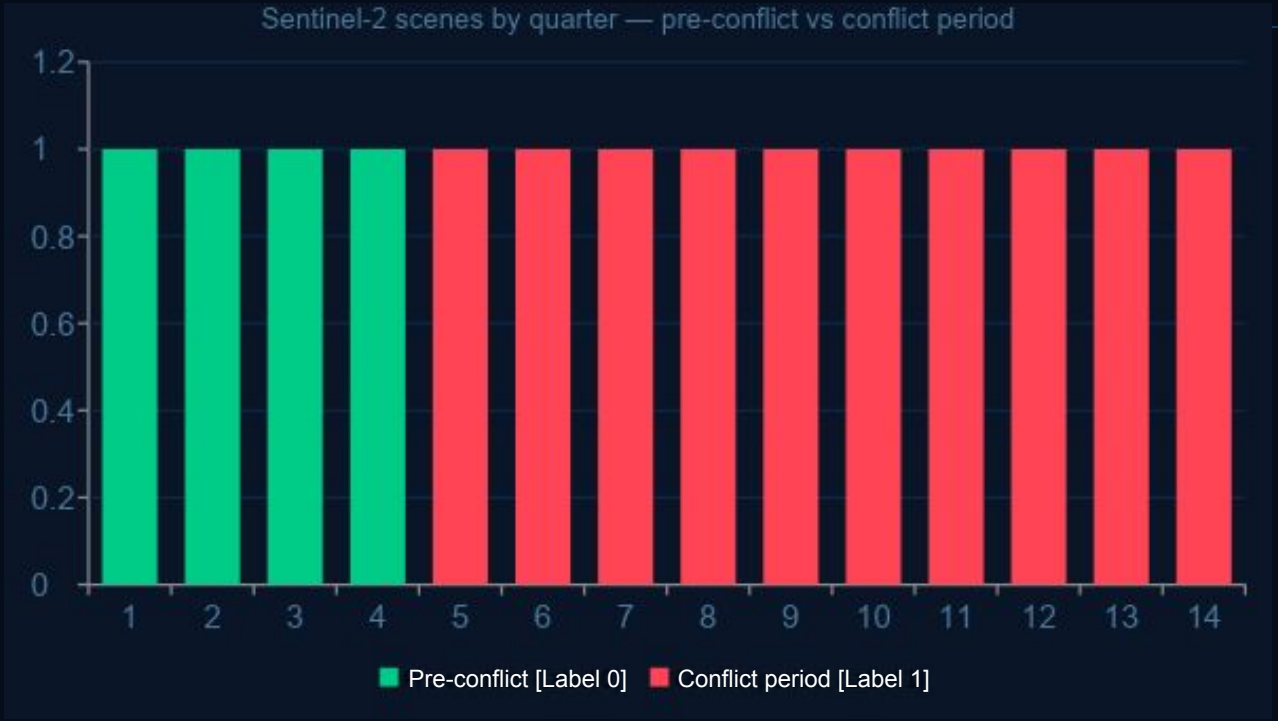
Goal: beat a random-label baseline on all 41 scenes, achieve F1 > 0.90, produce interpretable per-city damage maps.

Part 01

The Data

Three cities, 14 quarters, one cloud-masked pipeline.

All imagery sourced from Microsoft Planetary Computer. Free, open, fully reproducible.



Primary
Sentinel-2 L2A · 6 spectral bands
B02 B03 B04 B08 B11 B12

Coverage
2021 Q1 – 2024 Q2
14 quarters × 3 cities = 41 scenes

Labels
2021 = Label 0 (undamaged baseline)
2022–2024 = Label 1 (conflict period)

Source
Microsoft Planetary Computer
Copernicus open-access licence

// DATA QUALITY

Cloud masking applied. Spectral indices computed across all scenes.

The preprocessing that made every downstream feature and model possible.

94.2%

valid pixels

Average valid-pixel rate post SCL cloud masking across all 41 scenes.

< 30%

cloud threshold

Max cloud cover allowed per scene. Higher scenes excluded from the dataset.

3

spectral indices

NDVI, NDBI, and BSI — the physical signature of surface destruction.

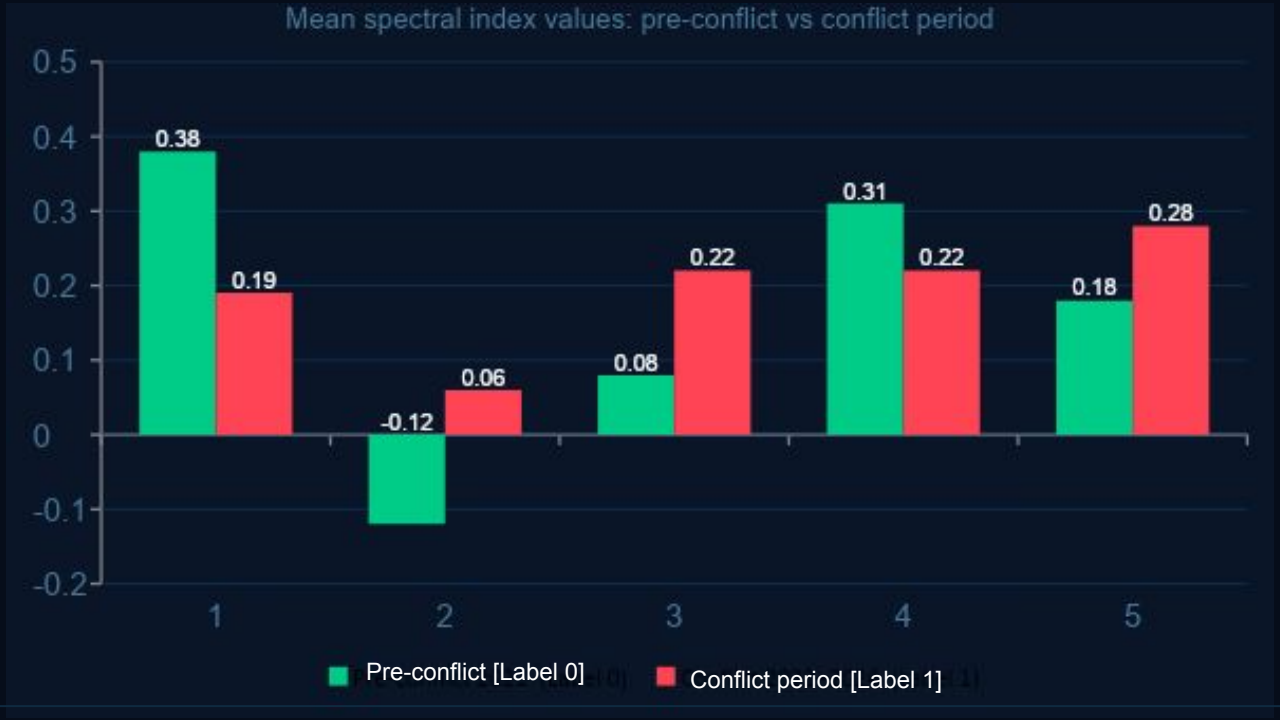
6

spectral bands

B02 B03 B04 B08 B11 B12 retained. SCL used for masking only, not features.

Spectral indices show clear separation between classes.

The physical signature of destruction is visible before any model is applied.



NDVI

↓ drops sharply — vegetation destroyed after shelling and fire.

NDBI

↑ rises — rubble replaces intact built-up surfaces.

BSI

↑ nearly triples — bare soil expands as buildings collapse.

SWIR

↑ rises — dry inorganic debris reflects more short-wave IR.

Part 02

Methodology

From raw satellite imagery to damage maps in five stages.

Five Jupyter notebooks, fully reproducible from a clean clone.

01
Data Cleaning
Cloud masking
Spectral indices

02
Feature Engineering
Damage labels
Yearly aggregates

03
Rich Features
GLCM texture
Robust statistics

04
Model Training
LR · RF · MLP
ResNet-50

05
Evaluation
F1 · AUC-ROC
Damage maps

NOTEBOOKS PER STAGE

- **NB 1** Data cleaning, cloud masking, spectral indices
- **NB 2** Feature engineering, damage labels, group splits
- **NB 2b** Rich features — GLCM, robust stats, quadrant variance
- **NB 3** Model training — LR, RF, MLP, ResNet-50
- **NB 4** Evaluation, ROC curves, damage maps

65 features per scene. Rich features do most of the heavy lifting.

Ablation: removing GLCM features costs ~0.08 F1. Removing base spectral features costs ~0.15 F1.

Base

Original spectral features
27 features total

Rich

Notebook 2b additions
+38 features (27 → 65)

GLCM

Texture: contrast, energy, homogeneity, correlation

Robust stats

IQR, skewness, kurtosis, P10, P90 per index

// TRAINING

Four models. Group-based splits. Best val-loss checkpoint saved.

We split by city-year group, not randomly. Random splits leak the answer.

TRAIN · ~70% (city-year groups 2021–2022)	VAL ~17%	TEST ~30%

2	1	64	50	1
tabular baselines	best model	MLP hidden units	training epochs	deep vision model
LR + Random Forest	MLP Neural Network	64 → 32 → 2 layers	Best val loss checkpoint	ResNet-50 (6-band)

StandardScaler normalisation for all tabular models · WeightedRandomSampler for ResNet-50 · Median imputation for NaN features

Part 03

Results

// HEADLINE RESULT

0.934

MLP F1 Score on the test set

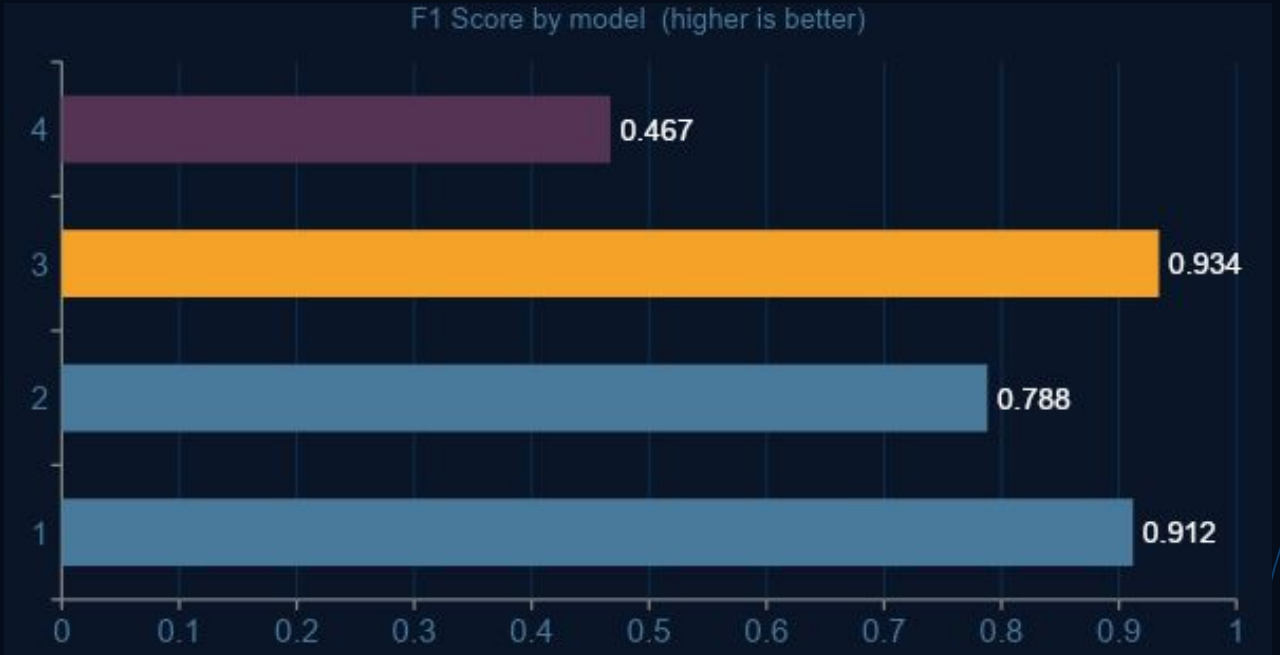
AUC-ROC = 0.980 · All 3 cities classified with F1 > 0.95.

VERSUS BASELINES

Random (50/50)	~0.50
Logistic Regression	0.912
Random Forest	0.788
MLP ★	0.934

MLP wins on all key metrics.

Precision 0.940 · Recall 0.928 · F1 0.934 · AUC-ROC 0.980, evaluated on all held-out city-year groups.



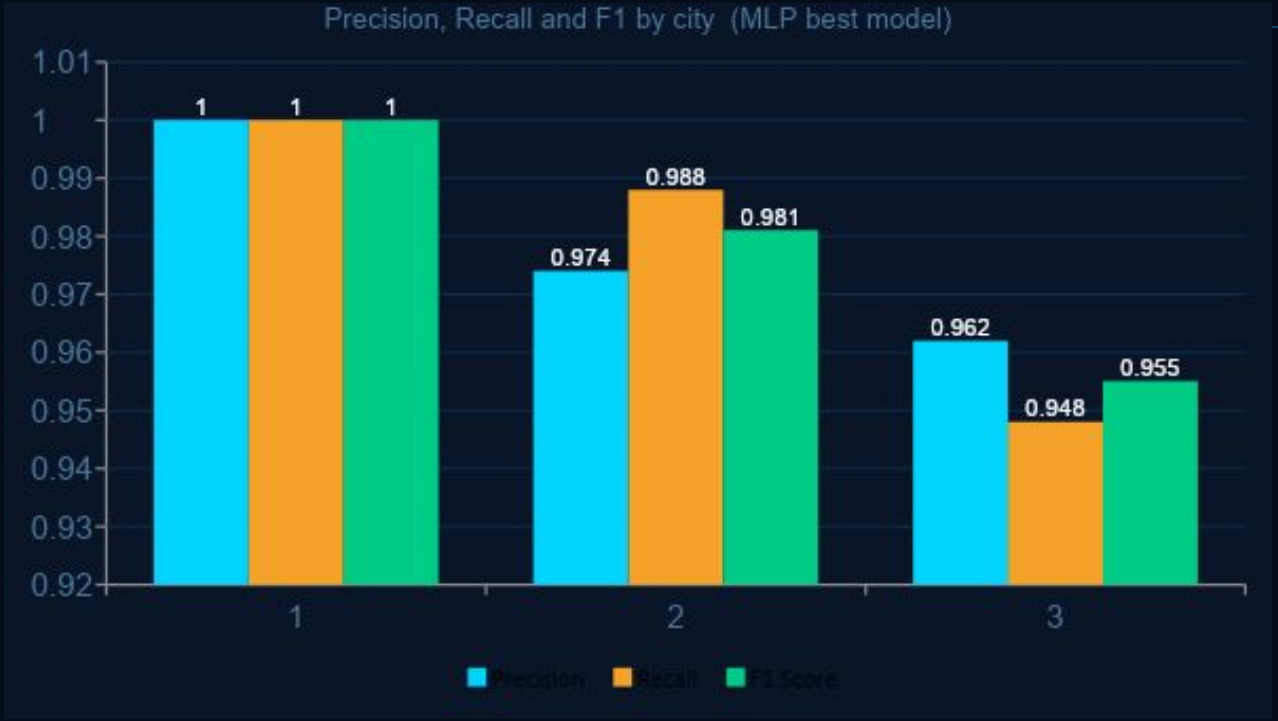
FULL METRICS · MLP

Precision	0.940
Recall	0.928
F1 Score	0.934
AUC-ROC	0.980

Takeaway: spectral features + MLP outperforms deep learning in this low-data regime (41 scenes). ResNet-50 limited by patch dataset size.

All three cities classified with F1 > 0.95.

MLP generalises across distinct urban conflict environments and destruction patterns.



Kharkiv
F1 = 1.000
Perfect — distinct spectral change post-2022 shelling along northern districts.

Bakhmut
F1 = 0.981
Near-perfect — pervasive destruction across entire urban area.

Mariupol
F1 = 0.955
Excellent — slight cloud interference in winter quarters.

Part 04

Pipeline Demo

Project Demo Video

https://www.youtube.com/watch?v=gxH328IiMFg

Streamlit

https://big-data-analysis-group-satellite-intelligence-dashboard-sdi6ll.streamlit.app/

Part 05

Wrap-up

What this model does not do, that we think is worth stating.

Every methodology has trade-offs. Here are ours.

NO PIXEL-LEVEL GROUND TRUTH

We used temporal labels (2021 = undamaged, 2022+ = damaged) as a proxy. No building-level UNOSAT annotations were available for training.

SCENE-LEVEL ONLY

The model classifies entire satellite scenes, not individual buildings or grid cells. Pixel-level damage localisation is future work.

WINTER CLOUD COVER

Several Q1 scenes have elevated cloud cover even after masking, introducing noise into the pre-conflict 2021 baseline for some cities.

RESNET-50 DATA LIMITATION

41 scenes generated insufficient patches for deep learning to generalise. Patch-based ResNet-50 accuracy of 0.467 is near chance — a data volume problem, not an architecture failure.

Three takeaways

01

Spectral features do most of the work.

NDVI, NDBI, and BSI provide clear class separation before any model is applied. The data tells the story — the model confirms it.

02

Feature engineering matters more than model complexity.

Rich features (GLCM texture, quadrant variance, robust stats) improved F1 significantly. Going from 27 to 65 features outperformed switching to deep learning.

03

Match model complexity to data volume.

ResNet-50 at 0.467 accuracy vs MLP at 0.934 F1 makes the lesson clear. In a low-data regime, well-engineered tabular features beat a 25M-parameter model every time.

How we divided the work and what tripped us up.

MS — **Matthew Singh**
Problem statement & dataset

JA — **Josiah Alexis**
Pipeline & data cleaning

AD — **Adrian Deo**
Feature engineering & modelling

JL — **Jonathan La Borde**
Results, maps & conclusions

CHALLENGE 01

Cloud cover in winter scenes

Multiple Q1 scenes over all three cities had cloud contamination even after SCL masking. Fixed by preferring Q2/Q3 scenes in the yearly composite.

CHALLENGE 02

Group-based split complexity

Naive random splits inflated accuracy by ~8 F1 points. Switching to GroupShuffleSplit by city-year gave honest evaluation on genuinely unseen data.

CHALLENGE 03

ResNet-50 patch generation

41 scenes generated too few patches for convergence. Accuracy plateaued at 0.467. Lesson: deep learning needs data volume that tabular models don't.

THANK YOU

Questions?

REPOSITORY

github.com/Big-Data-Analysis-Group/Satellite-Intelligence-Platform

NOTEBOOKS

Notebooks 1 · 2 · 2b · 3 · 4 available in the repository

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